Sam Kravitz

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WORK EXPERIENCE

American Thrombosis & Hemostasis Network (ATHN) **Nov. 2021 - Present**
Software Engineer *Remote*

- Designed and delivered scalable software systems for a Ruby on Rails clinical research platform supporting nationwide studies, owning architecture, implementation, and long-term evolution of core application infrastructure.
- Architected a reusable application framework that automated generation of 100+ configurable clinical workflows, significantly reducing implementation effort while improving consistency across studies.
- Designed the application's validation architecture, enforcing consistency across thousands of patient records and complex multi-step workflows while reducing opportunities for data integrity issues.
- Planned and executed large scale Ruby and Rails modernization initiatives, reducing technical debt while maintaining application stability throughout production upgrades.

Applied Research Associates **Jun. 2021 - Oct. 2021**
Software Engineer *Raleigh, NC*

- Designed and implemented high-performance C++ components for mission-critical systems requiring reliability, maintainability, and low-latency communication.
- Contributed to the full software development lifecycle, from requirements gathering and system design to testing and deployment, ensuring the delivery of effective and correct software aligned with client specifications.
- Collaborated with multidisciplinary teams to ensure solutions met strict client specifications and operational constraints.

SELECTED PROJECTS

Hinawa **C++**
Hobby Web Engine

- Built a browser engine from scratch in C++: HTML tokenizer/parser, CSS parser + selector engine, box-model layout engine, and a Skia-backed painter, rendering real pages pixel-for-pixel.
- Implemented a full JavaScript engine as an embeddable system, validated against a language conformance suite.

Sports Analytics Platform **Ruby/Go**

- Designed a data ingestion platform aggregating live sports data from multiple APIs into PostgresSQL.
- Built Go microservices responsible for scheduled data collection and normalization.
- Integrated AI-assisted analysis using retrieval-augmented generation (RAG) over structured sports data.

EDUCATION

Purdue University *West Lafayette, IN*
Bachelor of Science, Computer Science GPA: 3.6/4.0

SKILLS & INTERESTS

- **Languages:** Ruby, C/C++, Typescript/Javascript, Rust, Python
- **Systems:** System Architecture, Relational Data Modeling, Distributed Systems, Optimization, Developer Tooling
- **AI:** RAG, LLM Integration, Amazon Bedrock, AI Assisted Development
- **Interests:** Triathlon, Chess, Basketball, Video Game Development, Music Production